Reading · Part 4 — Energy

Multilevel · Questions 21–26

Energy is the power derived by using chemical or physical resources, and it is the strength and vitality required for sustained mental or physical activity. People and animals get energy from eating food. The plants receive their energy from the Sun, and in turn people eat the plants and animals that have also eaten plants – all made possible by the food chain.

Non-renewable energy sources are those that take millions of years to form and will run out some day. It is energy that comes from fossil fuels such as coal, crude oil, and natural gas. Fossil fuels are mainly made up of carbon and were formed millions of years ago. However, renewable energy sources will never run out, are better for the environment and do not cause pollution. They can sometimes cost a little more money. Renewable energy sources include the Sun, wind, water, heat from the earth, and biomass (plants).

Renewable energy is often called green energy because it is a natural energy, always available and does not have to be formed like non-renewable energy. The green energy is always there. For example, the Sun consistently shines, water is abundant, and the winds blow throughout the year. The five types include solar, energy from the Sun; geothermal, energy from heat within the Earth; hydroelectric, energy from moving water; biomass, energy from dead plants and microorganisms; and finally, energy from the wind.

For solar, the energy from the sun is captured in one of two ways. Active uses special technology and equipment to use the energy and focus the sunlight in a specific spot, generating electricity. Passive uses no equipment, uses the sunlight as it naturally changes throughout the day.

Windmills have been used since ancient times, to grind grain, power boats, or for pumping water. Today, wind turbines include tall towers and 2 or 3 propeller-like blades at the top that are turned by the wind. The blades turn a generator inside the tower to produce electricity. Groups of these turbines are called wind farms, found on farmland, in narrow mountain passes, or in the ocean.

Geothermal energy uses heat from the core of the Earth. The heat is always moving towards the surface. Underground rocks melt into magma and come to the surface as lava. Underground sources of water can shoot out as geysers. The sources can be accessed using geothermal heat pumps, bringing the heat aboveground to be used as energy. In some areas of the world, steam can be pumped directly to a power plant, produced by water heated underground.

Hydroelectric energy is made by flowing water as power plants usually located on large dams control the flow of the water, and as dams block a river, they create artificial lakes or reservoirs. Water from the lakes or reservoirs is forced through tunnels, and as it flows, it turns huge turbines to generate electricity. Niagara Falls in New York is an example of a place where hydroelectric energy is produced.

Biomass energy comes from the recently living plants or microorganisms. The energy in plants comes from the sun but is still present when it dies. Examples include trees, branches, scraps of bark and recycled paper, as well as manure, garbage, and some crops. The energy comes from burning the biomass. Some biomass can be converted into biofuels as it is mixed with regular gasoline.

As with both renewable and non-renewable energy sources, there are advantages and disadvantages. In the United States, as of 2016, about 10% of total energy consumption was from renewable energy sources, and about 55% of that use is for producing electricity. One of the most important advantages of renewable energy is its role in reducing greenhouse gases. The use of renewable energy in the U.S. and the world will continue to grow into the future.

Questions 21–26

Choose the correct answer (A, B, C or D).

21. Hoover Dam in the Southwest provides energy for millions of people. Which of the following types of renewable energy is being used?

A) Geothermal
B) Solar
C) Wind
D) Hydroelectric

22. Areas near volcanoes throughout the world may be sources for which type of energy?

A) Hydroelectric
B) Biomass
C) Geothermal
D) Solar

23. Which of the following energy sources includes trees, branches, scraps of bark and recycled paper, as well as manure, garbage, and some crops?

A) Geothermal
B) Biomass
C) Hydroelectric
D) Wind

24. Which of the following comes to the surface during the use of geothermal energy?

A) Lava
B) Magma
C) Geysers
D) All of the above

25. Renewable energy is often called which of the following?

A) Blue energy
B) Green energy
C) Red energy
D) Yellow energy

26. All the following are fossil fuels EXCEPT:

A) Coal
B) Crude oil
C) Solar
D) Natural gas

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Answer key

21-D 22-C 23-B 24-D 25-B 26-C

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